KEDGE Business School EBP1 Quantitative Methods 1

Worksheet 7 — Integrated cases

Exercise statements

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1 Assessment and exam practice

Practise independently with assessment and final-exam style questions.

Assessment / exam practice — 70 min

1.1 Case A – Continuous-assessment preparation

A school surveys 200 students about the maximum price they would accept for an optional service:

Price	$[5,10[$	$[10,15[$	$[15,20[$	$[20,30[$	$[30,40[$	$[40,60[$
n_i	10	25	45	60	40	20

Identify population, unit and variable type; build the complete table with widths, densities, frequencies, ICC, DCC and midpoints; draw histogram and cumulative curves; calculate modal class, median, quartiles, mean, population variance, standard deviation and CV; find the price affordable to about 65%; then classify a satisfied/not satisfied variable and choose a graph.

Assessment / exam practice — 110 min

1.2 Case B — New mock final exam — 110 minutes

Independent mock exam: the data below differ from exercises in chapters 4–6. Show every calculation step and justify rounding.

Part 1 — Two regressions (40 min)

t	1	2	3	4	5	6
x	5	6	8	9	11	13
y	12	15	18	20	24	29

1. Build t, x, y, t^2, x^2, y^2, tx, xy table; calculate both correlations and the regressions x on t and y on x. 2. Forecast x and y for t=8; qualify the forecast.

Part 2 — Quarterly series (40 min)

Année / Year	Q1	Q2	Q3	Q4
A	72	88	80	110
B	78	96	88	121

1. Plot the series; calculate MA4 and CMA4. 2. With the supplied trend T_t=2.8t+78, compute all eight ratios Y_t/T_t. 3. Average by quarter, normalise (sum 4) and forecast t=9 to 12.

Part 3 — Index numbers (30 min)

Produit / Product	p_0	q_0	p_1	q_1
A	10	40	12	45
B	8	60	9	55
C	20	20	22	25

Compute all four aggregates; Laspeyres, Paasche and Fisher price and quantity indices; the value index; and verify Fisher's factor reversal.